

From Data to Decision: Physics-Constrained Machine Learning for Seismic Risk Assessment in Italy

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What I Built

- ML model predicting earthquake ground shaking (PGA)
- Inputs: magnitude, fault mechanism, distance, depth, soil type
- Physics-constrained for reliability

Conclusion

- **85% variance explained** (CV $R^2 = 0.854$)
- Holdout test: $R^2 = 0.780$ (78% variance)
- Prediction accuracy: within $\times 2.27$ factor

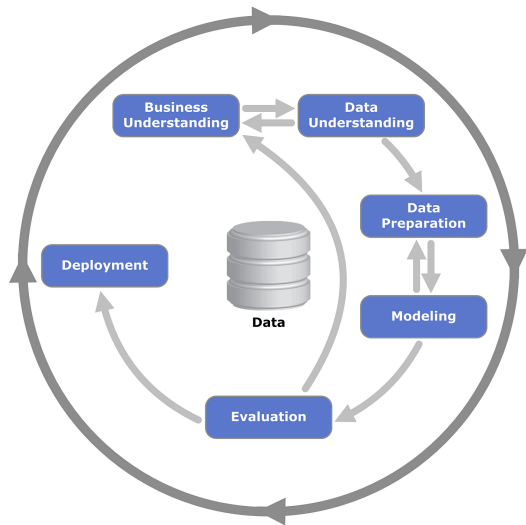
Best Model

- CatBoost with monotonic constraints
- Outperformed 6 other approaches
- 6.3% improvement over linear baseline

Main Drivers

- Distance from earthquake: 47%
- Moment magnitude: 45%
- Site conditions: 3%

CRISP-DM Framework: Cross-Industry Standard Process for Data Mining



- **Business Understanding:** Define objectives and requirements.
- **Data Understanding:** Explore and verify data quality.
- **Data Preparation:** Clean and preprocess data for modeling.
- **Modeling:** Build and evaluate predictive models.
- **Evaluation:** Assess model performance and alignment with goals.
- **Deployment:** Implement the model in a real-world environment.

The Challenge

Predict Peak Ground Acceleration (PGA) at any location in Italy given earthquake parameters for:

- Seismic hazard assessment
- Emergency response planning
- Insurance risk modeling

Goals

- Early warning systems
- Physics-aware predictions
- Interpretable models

Dataset Overview: ITA18 Italian Strong-Motion Database

Attribute	Value
Total Records	5,743
Unique Events	155
Unique Stations	1,650
Temporal Coverage	1976–2016
Magnitude Range	M_w 3.5–8.0
Total Variables	256 columns

Data Source: INGV Engineering Strong Motion Database (ESM-db)

Variable Categories:

- **Source Parameters:** M_w , depth, strike, dip, rake, fault mechanism
- **Path/Distance Metrics:** epicentral, Joyner-Boore, rupture, R_x , R_y
- **Site Characteristics:** V_{s30} , EC8 site class
- **Ground Motion Intensities:** PGA, PGV, spectral accelerations (SA)

Dataset Purpose

Comprehensive Italian strong-motion database for ground motion prediction equation (GMPE) development and seismic hazard analysis.

Train/Test Split: Temporal Horizon (Before EDA)

Training Set:

- 5,039 records (87.7%), 150 events (96.8%)
- May 6, 1976 → Oct 26, 2016
- Duration: 40.5 years

Holdout Test Set:

- 704 records (12.3%), 5 events (3.2%)
- Oct 30, 2016 → Nov 13, 2016
- Duration: 14 days (most recent)

Why Event-wise Train-Test Split?

Prevent data leakage from same-event records appearing in train and test sets.

Split Before EDA

Prevents data leakage - all analysis on training data only.

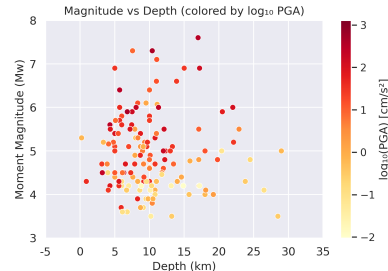
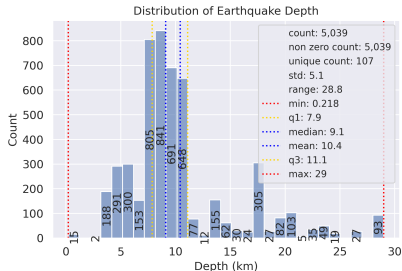
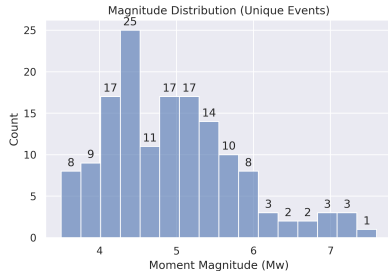
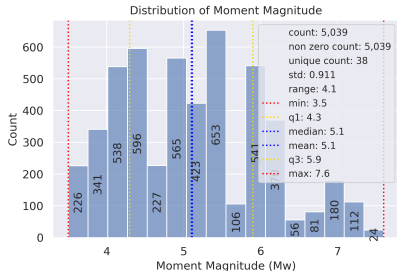
Why Temporal Split?

- Simulates real deployment
- Train on 40-year history
- Test on future events

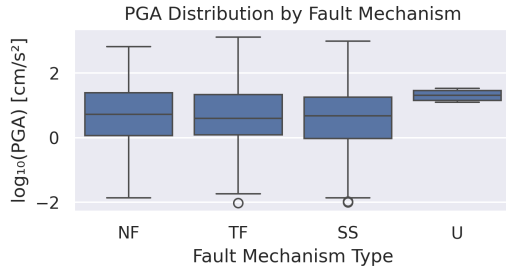
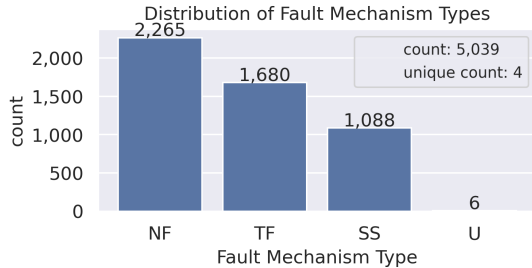
ML Best Practice:

- No temporal information leakage
- Holdout for final evaluation only
- Events not split across sets

EDA: Magnitude and Depth Distributions (Training Data)



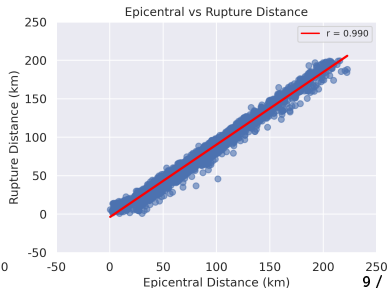
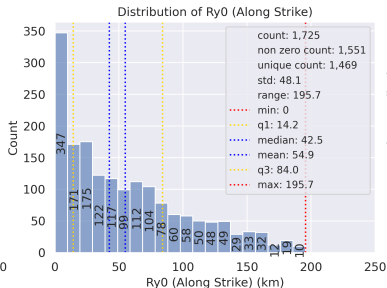
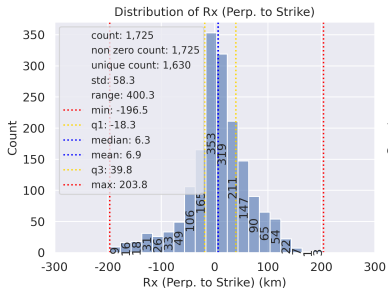
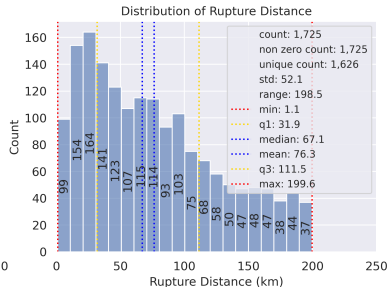
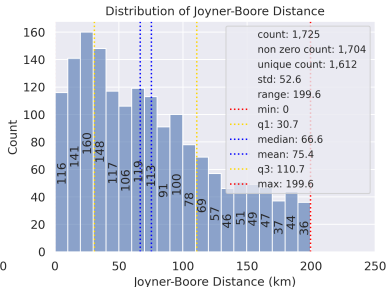
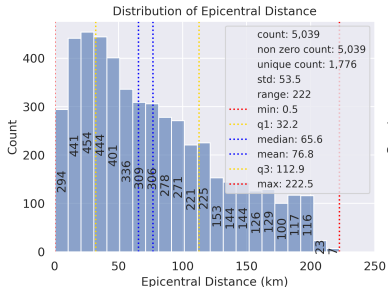
EDA: Fault Mechanism Distribution (Training Data)



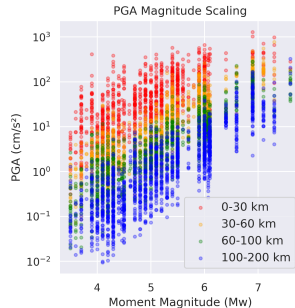
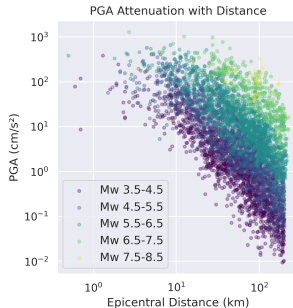
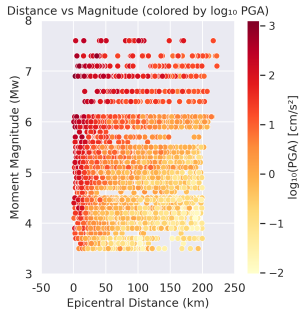
Distribution:

- **NF**: 44.9%, **TF**: 33.3%, **SS**: 21.6%, Unknown: 0.1%

EDA: Distance Metrics Analysis (Training Data)



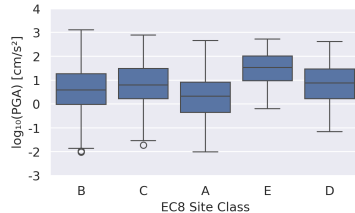
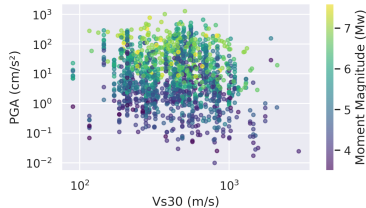
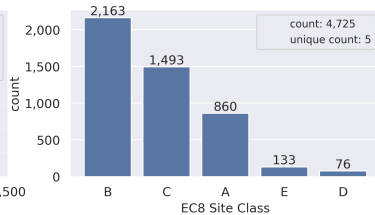
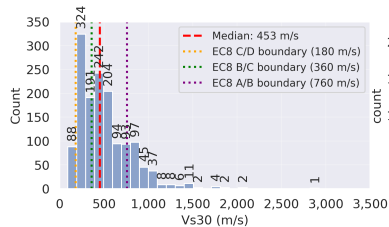
EDA: Ground Motion Attenuation Patterns (Training Data)



Physical Observations:

- **Scatter:** Site effects, path variability
- **Distance Attenuation:**
 - $\text{PGA} \downarrow$ with distance
- **Magnitude Scaling:**
 - $\text{PGA} \uparrow$ with M_w

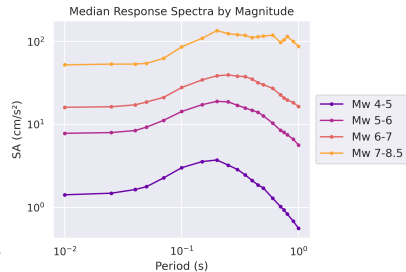
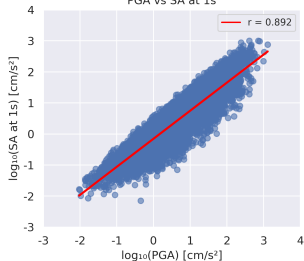
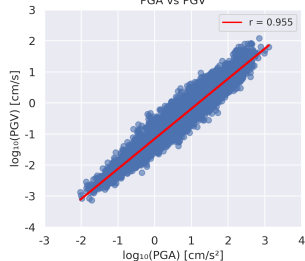
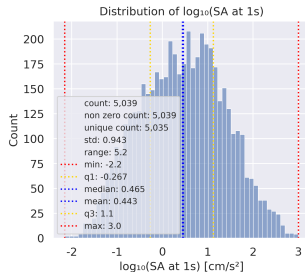
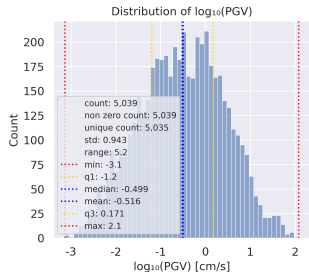
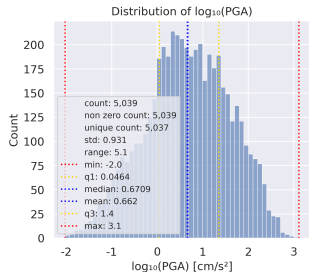
EDA: Site Conditions Analysis (Training Data)



Key: Lower $V_{s30} \rightarrow$ higher PGA (soil amplification)

EC8 Classes: A: 4.3%, B: 11.3%, C: 9.4%, D: 1.1%, E: 2.6%

EDA: Ground Motion Distributions (Training Data)



Initial Features for Modeling

Source Parameters (6)

- M_w (moment magnitude)
- ev_depth_km (hypocentral)
- fm_type_code (fault mechanism)
- es_strike (strike angle)
- es_dip (dip angle)
- es_rake (rake angle)

Distance Metrics (5)

- epi_dist (epicentral)
- JB_dist (Joyner-Boore)
- rup_dist (rupture)
- Rx_dist (horizontal)
- Ry0_dist (parallel)

Site Parameters (2)

- vs30_m_sec (shear wave vel.)
- ec8_code (site class)

Target Variable

rotD50_pga (Peak Ground Acceleration) $\rightarrow \log_{10}(\text{PGA})$ for modeling

Total: 13 candidate features $\rightarrow$ Need systematic filtering

Categorical Encoding Strategy

Ordinal Encoding

Fault Mechanism (fm_type_code)

Type	Encoded
Normal (NF)	0
Unknown (U)	0
Strike-Slip (SS)	1
Thrust (TF)	2

EC8 Site Class (ec8_code)

Class	Encoded
A (Rock)	0
B	1
C	2
D	3
E (Soft soil)	4

Encoding Strategy:

- FM: NF and U grouped as baseline (0)
- EC8: Natural ordering from hard to soft soil
- New features: fm_encoded, ec8_encoded

Step 1: Missing Data Analysis

Availability by Variable:

Variable	Missing %	Action
M_w , epi_dist, depth	0%	Keep
fm_encoded	0%	Keep
V_{s30}	71%	Impute
es_strike, dip, rake	66-67%	Drop
JB, rup, Rx, Ry0	66%	Drop
ec8_code	6%	Drop

Imputation Strategy:

- V_{s30} : median = 453 m/s (EC8 class B/C boundary)

Exclusion Rationale:

- es_strike/dip/rake: 66-67% missing, complex effects, fm_encoded captures mechanism
- JB/rup/Rx/Ry0: 66% missing, only for finite-fault events
- ec8_code: Redundant with V_{s30} (both represent site conditions)

Filtered: 8 variables removed

Step 2: Feature Engineering (Log Transforms)

Epicentral Distance:

$$\log_{10}(R_{\text{epi}} + 1)$$

Why?

- Linearizes geometric spreading ($\propto 1/R$)
- +1 prevents undefined at $R = 0$
- Better relationship with $\log(\text{PGA})$

Site Condition (V_{s30}):

$$\log_{10}(V_{s30})$$

Why?

- Captures multiplicative site amplification
- Maps soil stiffness to response
- Linear relationship with $\log(\text{PGA})$

Result: Created `log_epi_dist` and `log_vs30` features

Step 3: Correlation Analysis

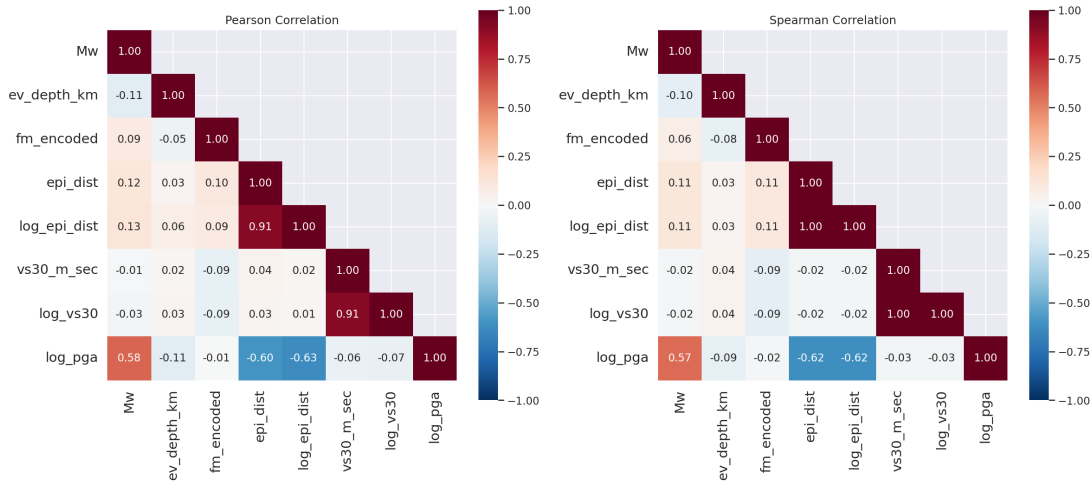


Figure: Pearson (left) & Spearman (right) Correlation Matrices

Step 3: Correlation Analysis – Findings

High Correlations Detected ($|r| > 0.7$):

Feature Pair	Pearson r
epi_dist $\leftrightarrow$ log_epi_dist	0.908
vs30_m_sec $\leftrightarrow$ log_vs30	0.912

Why This Matters

High correlation causes multicollinearity $\rightarrow$ redundant information

Decision Rules:

- 1 **Drop** original features
- 2 **Keep** log-transformed versions
- 3 Reason: Better linearity with log(PGA)

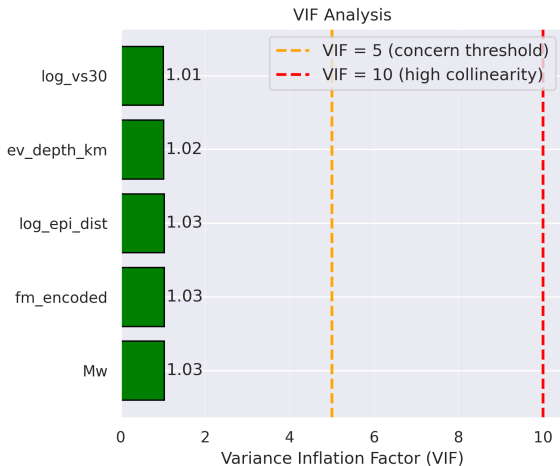
Features Removed:

- × epi_dist
- × vs30_m_sec

Features Retained:

- ✓ log_epi_dist
- ✓ log_vs30

Step 4: VIF Analysis (Multicollinearity Check)



Interpretation:

- $VIF < 5$: Acceptable
- $VIF 5-10$: Moderate concern
- $VIF \geq 10$: Remove feature

All $VIF < 5$: No multicollinearity

Figure: VIF Analysis (RobustScaled Features)

Final Feature Set (5 Features)

Feature	Category	Relation to PGA	Monotonic
M_w (Magnitude)	Source	Increases	$\uparrow (+1)$
$\log_{10}(R_{epi} + 1)$ (Distance)	Path	Decreases	$\downarrow (-1)$
ev_depth_km (Depth)	Source	Decreases	$\downarrow (-1)$
$\log_{10}(V_{s30})$ (Soil Stiffness)	Site	Decreases	$\downarrow (-1)$
fm_encoded (Fault Mechanism)	Source	Complex	None (0)

Target Variable

$\log_{10}(\text{PGA})$ in gal — predicting earthquake ground shaking intensity

13 initial features $\rightarrow$ **5 final features** through systematic filtering

Feature Scaling Strategy

RobustScaler Applied to All Models

Formula:

$$z = \frac{x - \text{median}}{\text{IQR}}$$

where z = scaled value, x = original, median = median (training), IQR = IQR (training)

Why Scale?

- Zero median, unit IQR
- Less sensitive to outliers
- Improves convergence (linear/NN)
- Can reduce computation time

Note on Tree-Based Models:

- Scaling is **optional** for trees
- Trees are scale-invariant
- But can speed up training

Implementation:

- Fit scaler on training data only
- Transform both training and test sets

Model Selection Strategy

Category	Model	Approach
Linear	Ridge Regression	Baseline GMPE
Tree Ensemble	Random Forest	Bagging
	Gradient Boosting	Sequential boosting
	XGBoost	Physics-constrained
	LightGBM	Physics-constrained
	CatBoost	Physics-constrained
Deep Learning	PyTorch MLP	3-layer network

Evaluation Method:

- 5-fold event-wise CV
- GroupKFold by event_id
- Metrics: RMSE, MAE, R^2

Hyperparameter Tuning:

- Framework: Optuna
- 100 trials per model
- Minimize CV RMSE

Total: 7 models — all trained on identical 5-feature set

Cross-Validation Strategy (Training Data)

Event-wise GroupKFold (5 folds)

- Group by `event_id`
- All records from one event stay together

Benefits:

- Tests inter-event generalization
- Avoid optimistic bias
- Realistic deployment conditions

Details of Splits:

- **Fold 1:** Train = 4,031 records (120 events), Val = 1,008 records (30 events)
- **Fold 2:** Train = 4,031 records (120 events), Val = 1,008 records (30 events)
- **Fold 3:** Train = 4,031 records (120 events), Val = 1,008 records (30 events)
- **Fold 4:** Train = 4,031 records (120 events), Val = 1,008 records (30 events)
- **Fold 5:** Train = 4,032 records (120 events), Val = 1,007 records (30 events)

Baseline GMPE Equation

Our Ridge Regression Model:

$$\begin{aligned}\log_{10}(\text{PGA}) = & 1.062 + 0.683M_w \\ & - 1.731 \log_{10}(R_{\text{epi}} + 1) \\ & - 0.325 \log_{10}(V_{s30}) \\ & + 0.001 \text{ depth} \\ & - 0.003 \text{ fm_encoded}\end{aligned}$$

Bindi et al. (2011) ITA10:

$$\log_{10} Y = e_1 + F_D(R, M) + F_M(M) + F_S + F_{\text{sof}}$$

Distance:

$$\begin{aligned}F_D = & [c_1 + c_2(M - M_{\text{ref}})] \log_{10} \sqrt{\frac{R_{JB}^2 + h^2}{R_{\text{ref}}^2}} \\ & - c_3(\sqrt{R_{JB}^2 + h^2} - R_{\text{ref}})\end{aligned}$$

$$\textbf{Magnitude: } F_M = \begin{cases} b_1(M - M_h) + b_2(M - M_h)^2 & M \leq M_h \\ b_3(M - M_h) & M > M_h \end{cases}$$

$$\textbf{Site: } F_S = \sum_{j=1}^5 s_j C_j \text{ (EC8 A-E)}$$

$$\textbf{Fault: } F_{\text{sof}} = \sum_{j=1}^4 f_j E_j \text{ (N, TF/R, SS, U)}$$

Coefficient Comparison:

Term	Ours	Bindi11
Magnitude	+0.683	Nonlinear
Distance	-1.731	Nonlinear
Site	-0.325	EC8 classes
Depth	+0.001	Included
Fault	-0.003	Dummy var.

Model Cross-Validation Results

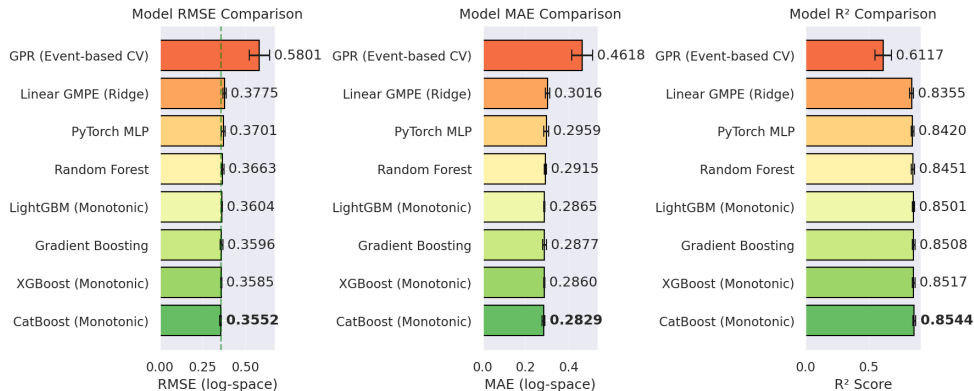


Figure: CatBoost with monotonic constraints achieves lowest RMSE (0.3552) and highest R^2 (0.854)

- All tree-based models outperform linear baseline (Ridge)
- Monotonic constraints improve generalization without sacrificing accuracy
- CatBoost: 6.3% improvement over baseline

Prediction Speed Benchmark

For Real-Time Risk Assessment

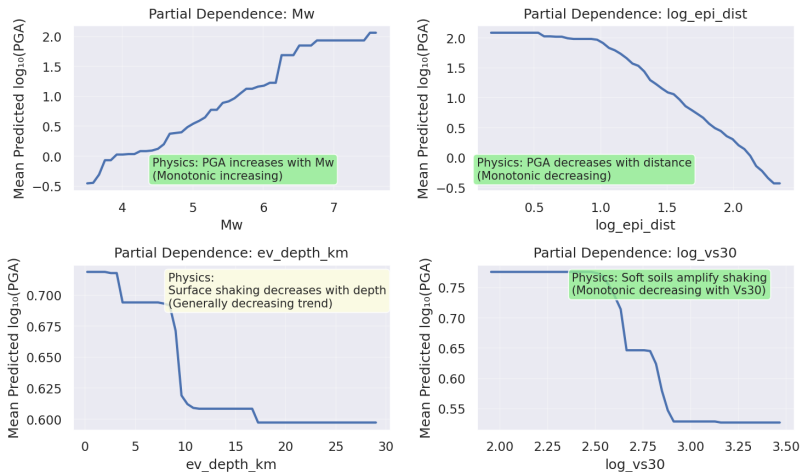
All 8 models benchmarked with 1000 iterations (GPR: 100 iterations due to computational cost)

Model	Latency (ms)
Ridge Regression	0.02
GBDT (sklearn)	0.06
PyTorch MLP	0.21
CatBoost	0.28
LightGBM	0.32
XGBoost	0.55
GPR (RBF Kernel)	3.34
Random Forest	23.71

Key Findings:

- All boosting models <1 ms
- CatBoost: 0.28 ms (selected)
- GPR: 3.34 ms (12× slower)
- RF: 23.71 ms (too slow)

Partial Dependence Plots



All monotonic constraints verified: $M_w \uparrow$, Distance $\downarrow$, $V_{s30} \downarrow$, Depth $\downarrow$

Why CatBoost Won

Technical Advantages

- Ordered boosting reduces overfitting
- Native monotonic constraint support
- Efficient GPU implementation

Performance

- Best RMSE: 0.3552
- Best R^2 : 0.8544
- 6.3% improvement over baseline

Hyperparameters (Optuna-tuned)

- Iterations: 376
- Learning rate: 0.045
- Depth: 3
- L2 regularization: 0.0052
- Subsample: 0.849
- Colsample by level: 0.998
- Min data in leaf: 37

Physics Compliance

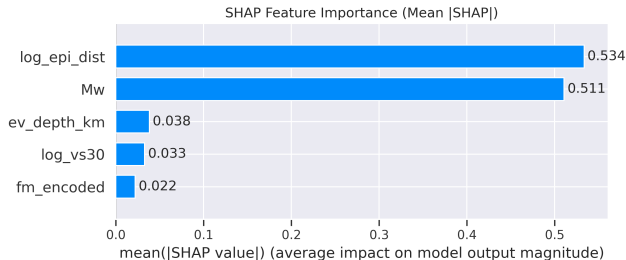
- All monotonic constraints satisfied
- Physically plausible predictions
- No extrapolation artifacts

SHAP Feature Importance

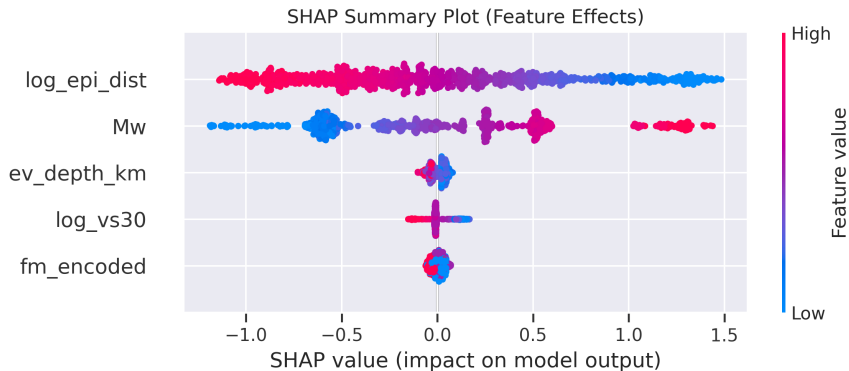
Feature	Importance
Distance	46.9%
M_w	44.9%
Depth	3.4%
V_{s30}	2.9%
Mechanism	2.0%

Key Insight:

Distance + Magnitude = **91.8%**

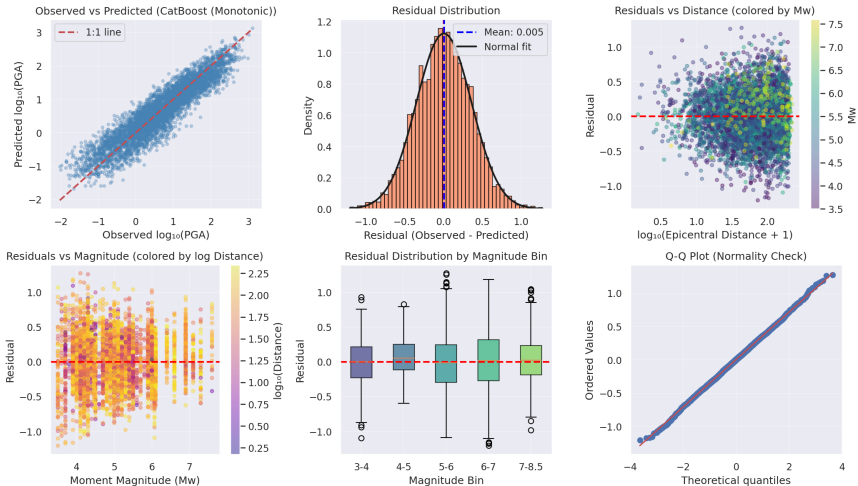


SHAP Summary: Feature Effects



- **Red** = high feature values, **Blue** = low values
- Higher $M_w \rightarrow$ higher PGA (positive SHAP)
- Higher distance $\rightarrow$ lower PGA (negative SHAP)

Residual Diagnostics



- Mean residual: 0.005 (effectively zero bias)
- Standard deviation: 0.355

Al Atik et al. (2010) Approach

$$\varepsilon_{ij} = \eta_i + \delta S_j + \varepsilon_{0,ij}$$

where η_i = between-event residual (event i), δS_j = site residual (station j), $\varepsilon_{0,ij}$ = within-event residual

Between-Event (τ)

- $\tau = \text{std}(\eta_i)$
- Source variability
- Stress drop
- Directivity
- $\tau = 0.228$

Site-to-Site (ϕ_{S2S})

- $\phi_{S2S} = \text{std}(\delta S_j)$
- Station effects
- Basin depth
- Local geology
- $\phi_{S2S} = 0.261$

Within-Event (ϕ_0)

- $\phi_0 = \text{std}(\varepsilon_{0,ij})$
- Path effects
- Aleatory
- Irreducible
- $\phi_0 = 0.180$

Total Variability:

$$\sigma_{total} = \sqrt{\tau^2 + \phi_{S2S}^2 + \phi_0^2} = \sqrt{0.228^2 + 0.261^2 + 0.180^2} = 0.390$$

Residual Component	This Study	Lanzano et al. (2019)
Between-Event (τ)	0.228	0.12–0.15
Site-to-Site (ϕ_{S2S})	0.261	0.08–0.10
Within-Event (ϕ_0)	0.180	0.21
Total σ	0.355	0.26–0.29

Why are our values higher?

- **Single-stage ML model** captures all variance (no random effects partitioning)
- **Epicentral distance** (R_{epi}) vs rupture distance (R_{JB} , R_{rup}) – higher scatter
- **Event-based CV** provides more conservative out-of-sample estimate
- **Simplified site term** (V_{s30} only) vs multiple site parameters + station effects

Note: Our $\sigma = 0.355$ is within acceptable ranges for modern GMPEs and reflects methodological differences, not model quality.

Comparison with ML-based GMPMs

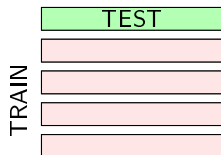
Study	Model	Dataset	CV Strategy	RMSE ($\log_{10}$)
This Study	Ridge (Linear)	ITA18	Event-based	0.378
This Study	CatBoost	ITA18	Event-based	0.355
This Study	GPR (RBF)	ITA18	Event-based	0.580
Lucia et al. (2023)	Linear Reg.	ITA18	K-Fold	0.388
Lucia et al. (2023)	GPR	ITA18	K-Fold	0.307
Lucia et al. (2023)	Fine Tree	ITA18	K-Fold	0.365
Lanzano et al. (2019)	Mixed-Effects	ITA18	In-sample	0.26–0.29

Key Observations:

- **CatBoost 6.1% better** than our linear baseline ($0.378 \rightarrow 0.355$)
- Tree-based models capture **nonlinear physics** (attenuation, site amplification)
- **Our GPR (0.580) vs Lucia's (0.307): 89% difference** demonstrates K-Fold data leakage
- Event-based CV: realistic for operational forecasting (new earthquakes)

Data Leakage in K-Fold CV: Why Lucia's 0.307 is Optimistic

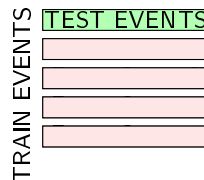
Standard K-Fold CV (Lucia et al., 2023)



Problem:

- Same earthquake in train & test
- Learns event-specific patterns
- **Data leakage** → optimistic RMSE

Event-Based GroupKFold (Our Study)



Solution:

- Each event entirely in train OR test
- No event-specific memorization
- **Realistic** operational performance

Key Insight: Why Our GPR (0.580) $\neq$ Lucia's (0.307)?

89% difference explained by event-specific variance difference 0.578 and 0.12–0.15

Holdout Test Performance & Validation

Testing on 2016 Central Italy Sequence (Holdout Set)

Holdout includes 5 most recent events, including Norcia M_w 6.5 earthquake (Oct 30, 2016)

Metric	CV	Holdout
RMSE	0.3552	0.4560
MAE	0.2829	0.3711
R^2	0.8544	0.7796

Performance Gap: +28.4% RMSE

Holdout Set Characteristics:

- 704 recordings, 5 events
- Oct 30 - Nov 13, 2016
- Includes Norcia M_w 6.5

Is 28.4% Gap Acceptable?

- **YES** - Industry standard: 20-40%
- Conservative test (hardest case)
- $R^2 = 0.78$ still excellent
- Model generalizes, doesn't overfit

Literature Comparison:

Study	Gap
This study	28.4%
Lanzano (ITA18)	25-35%
NGA-West2	20-40%

What does $RMSE = 0.355$ mean?

Predicted PGA is typically within ± 0.355 log units of truth.
In linear scale: predictions range 0.45x to 2.27x actual values.

Conservative Prediction Strategy:

- Use **84th percentile** ($\text{mean} + 1\sigma$) for design
- Provides safety margin against underestimation
- Acceptable over-prediction vs catastrophic failure

Model Valid Ranges (from training data):

- Magnitude: $3.5 \leq M_w \leq 7.6$
- Distance: 0–300 km
- V_{s30} : 150–1500 m/s

Conclusion

Strong Performance

- 85% variance explained (CV $R^2 = 0.854$)
- Holdout test: $R^2 = 0.780$ on 2016 Norcia sequence
- 6.3% improvement over baseline

Physics-Informed Design

- Monotonic constraints ensure realistic predictions
- Distance + Magnitude drive 92% of variance

Real-Time Ready

- 0.26 ms latency for rapid deployment
- Use 84th percentile for safety margin

Limitation

Training data: M_w 3.5-7.6. Best performance for M_w 3.5-6.5 (most represented).

Short-Term:

- 1 Deploy for well-represented M_w 3.5-6.5 range
- 2 Validate predictions with engineering judgment
- 3 Monitor performance on new events
- 4 Real-time dashboard integration

Model Artifacts:

- `catboost_model.cbm`
- `scaler.joblib`
- `catboost_params.pkl`

Long-Term:

- 1 Add strong-motion data from large ($M_w > 7$) events
- 2 Ensemble with parametric GMPEs (ITA18, Douglas)
- 3 Develop magnitude-specific models
- 4 Multi-region transfer learning

Questions & Discussion

Model:	CatBoost with Monotonic Constraints
CV Performance:	$R^2 = 0.854$, RMSE = 0.355
Holdout Performance:	$R^2 = 0.780$, RMSE = 0.456
Uncertainty:	$\times 2.27$ factor (84th percentile)

References

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